



Towards Robust Speaker Recognition against Intrinsic Variation with Foundation Model Few-shot Tuning and Effective Speech Synthesis

Zhiyong Chen^{*14}, Shuhang Wu^{*1}, Xinnuo Li², Zhiqi Ai¹, Shugong Xu³

¹Shanghai University, China

²New York University, USA

³Xi'an Jiaotong Liverpool University, China

⁴Hithink RoyalFlush AI Research Institute, China

{zhiyongchen, shuhangwu, aizhiqi-work}@shu.edu.cn, shugong.xu@xjtlu.edu.cn

Abstract

Speaker recognition is essential for secure authentication and personalized voice assistants in smart home settings, but it faces challenges due to intrinsic speaker variability, such as aging and emotional fluctuations. Existing methods often rely on pretraining and require extensive data. To address these challenges, we propose a framework for time-varying and emotion-robust open-set identification (OSI) for smart home environments, utilizing few-shot foundation enrollment-time tuning and style-rich zero-shot text-to-speech (TTS) systems. We explore best practices for synthetic data selection and suitable open-set outlier-focused loss functions. Our proposed method improves handling emotional and aging variations in target speakers, enhancing robustness to intrinsic variability while maintaining resilience to unknown outliers. Experiments demonstrate strong generalization across multiple time-varying and emotionally rich benchmarks.

Index Terms: speaker recognition, speaker identification, few-shot learning, speech synthesis, robust recognition

1. Introduction

Speaker recognition is a crucial task that identifies a speaker from their voice, playing a vital role in secure authentication systems and personalized voice assistants in smart home. However, achieving robust performance in real-world scenarios remains challenging due to the inherent variability in speaker voice prints. While environmental noise and mismatched recording devices are known to degrade system performance, intrinsic speaker variations—such as aging and emotional fluctuations—can be just as disruptive. As speakers undergo physiological changes or emotional shifts, their voice characteristics (e.g., pitch, energy, prosody) change over time, undermining recognition accuracy. In some cases, repeated enrollment under different conditions has been used to mitigate these issues [1], but such approaches are often complicated and do not fully address the underlying variability.

Most existing studies on speaker recognition focus on enhancing robustness against device and environmental variability, often relying on large-scale pretraining with diverse speakers [2]. However, comparatively less attention has been given to managing intrinsic speaker variability, especially over time for a *specific group of targets*, which is particularly useful in smart home settings [3] (or open-set speaker identification OSI). Voice changes related to aging or emotional states remain a significant hurdle for short-term, continuous, or time-varying recognition tasks in these settings, where the goal is to robustly recognize all speakers despite variations. Existing methods for

age-invariant [4] or emotion-invariant representations often require extensive enrollment data and intensive training, making them impractical for real-world applications.

To address these challenges, we propose a novel framework for time-varying and emotion-robust speaker recognition during the enrollment stage using few-shot fine-tuning for target speakers. Rather than relying on exhaustive domain-specific pretraining, our approach uses speaker-specific zero-shot text-to-speech (TTS) systems to generate synthetic speech that captures the target speaker's varying emotional states and intrinsic vocal characteristics. This is particularly beneficial for groups of target speakers in enrollment scenarios. We then perform few-shot open-set fine-tuning with text-independent enrollment data, reducing the number of required utterances to as few as three, and explore the best practices for utilizing this augmented data. Through this process, we significantly enhance the system's robustness to intrinsic speaker variability.

Our major contributions are as follows:

- We demonstrate that recent zero-shot timbre-controllable speech synthesis techniques, combined with a synthetic data sampling control policy and training methodology, can effectively address intrinsic speaker variations and achieve robust speaker recognition in smart home OSI settings.
- We propose an open-set enrollment fine-tuning strategy that integrates synthetic and real data to improve speaker recognition OSI performance. This approach requires only limited enrollment samples while remaining focused on the target speaker and robust to outliers.
- We thoroughly evaluate our proposed method across several challenging tasks, including time-varying and emotionally rich evaluation benchmarks, showcasing its generalization capabilities in real-world speaker recognition scenarios.

2. Related Work

Recent research has largely focused on learning intrinsic-invariant speaker representations. For instance, [5, 6] introduced an emotion-robust framework that utilizes pretrained emotion-specific extractors and pretraining techniques for emotion-robust embeddings. For aging-robust speaker recognition, methods such as those in [1, 4, 7, 8] apply cross-aging learning techniques to improve speaker recognition performance over time. Most existing methods primarily focus on pretraining with cross-domain data for robustness or using template updating as a remedy for shifts in speaker characteristics.

Several studies have also explored the use of speech synthesis to strengthen speaker recognition. Previous research has examined data augmentation for text-dependent verification [9, 10] and accent recognition pretraining [11]. Earlier TTS sys-

^{*}Contributed equally to this work.

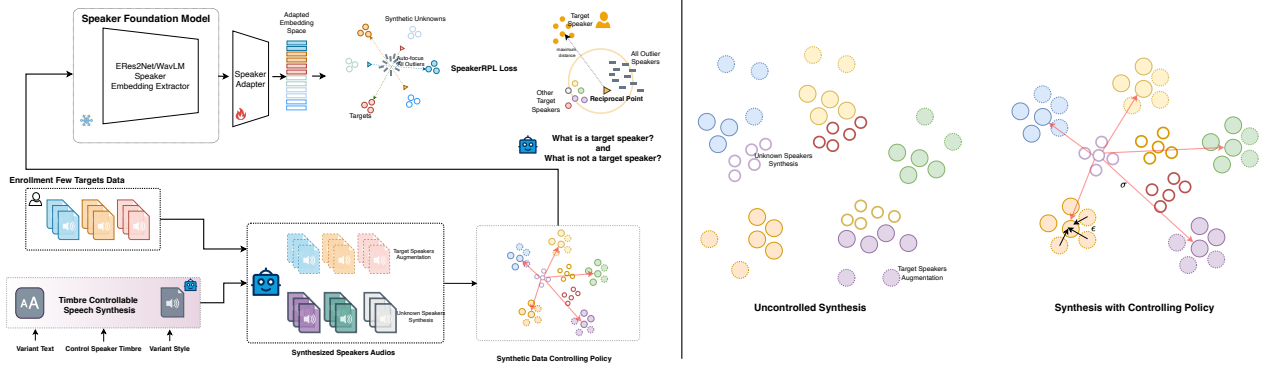


Figure 1: Overall architecture (left) and controlling policy for synthetic data (right).

tems have been applied to text-independent d-vector pretraining [12]. Additionally, parallel research in Spoofing-Aware Speaker Verification (SASV) [13, 14] has investigated the use of older-generation TTS systems, although these systems typically suffer from limited speech quality and style richness [15, 16].

While much work has been done on open-set identification [17, 18] and few-shot learning for speaker identification [19, 20], [21] investigates few-shot enrollment-time open-set learning on foundation models. However, this work is limited to *text-dependent scenarios and does not fully explore the challenges of intrinsic speaker variability*. In this work, we delve deeper into text-independent speaker recognition, utilizing effective and generalized foundation models, an optimal control policy for synthetic data, and a comprehensive comparison of open-set training losses to establish best practices.

3. Methods

3.1. Robust speaker recognition via speech synthesis and enrollment-time few-shot tuning

In our framework, we utilize both the enrollment dataset, D_{enroll} , and synthetic data, D_{syn} , generated by zero-shot TTS for target speakers and perform enrollment-time tuning (see Figure 1). We construct an enhanced enrollment set $D_{enroll+} = D_{enroll} \cup D_{syn}$, and optimize the open-set identification (OSI) recognition risk functions [22] to improve performance over the baseline foundation models. Specifically, we define the recognition risk as

$$\mathcal{R}(f_\theta) = \mathcal{R}_{target}(f_\theta, D_{test}^{targets}) + \mathcal{R}_o(f_\theta, D_{test}^{outliers}), \quad (1)$$

where $\mathcal{R}_{target}(\cdot)$ is the risk computed over target speakers, $\mathcal{R}_o(\cdot)$ is the open-set risk computed over outlier samples, and $D_{test}^{targets}$ and $D_{test}^{outliers}$ denote the target and outlier test sets, respectively. f_θ is the learnable speaker modeling function. Our objective is to minimize the recognition risk conditioned on the enhanced enrollment set: $f_\theta^* = \arg \min_{f_\theta} \mathcal{R}(f_\theta | D_{enroll+})$.

To adapt the original foundation models for speaker recognition, we employ a post-adaptor with learnable θ during training:

$$emb := F_{adapted}(x) = \text{Adapter}_\theta(F(x)), \quad (2)$$

where $F(\cdot)$ and $F_{adapted}(\cdot)$ denote the original and adapted embeddings, respectively.

3.2. Synthetic data selection with controlling policy

To effectively leverage the proposed synthetic data, a well-designed controlling policy for selecting D_{syn} is essential. We propose two key sampling rules, as illustrated in Figure 1 (right).

For each target speaker, we select synthetic samples that closely match the corresponding enrollment data. Specifically, for speaker k , we define the synthetic data as $D_{syn}^k = \{x_{syn}^k | \text{dist}(x_{syn}^k, x_{enroll}^k) < \epsilon\}$, where ϵ is a threshold that ensures the selected synthetic samples are sufficiently similar to any of speaker k 's enrollment data x_{enroll}^k . This strategy ensures that the speaker model is well-adapted to the target speaker's characteristics. In the implementation, we use a nearest-N method for this selection.

In addition to target control, we propose a method for selecting synthetic samples that emphasize the non-target, unknown speakers, which is crucial for improving performance. We construct synthetic samples for a new speaker k' as $D_{syn}^{k'} = \{x_{syn}^{k'} | \text{dist}(x_{syn}^{k'}, x_{enroll}^k) > \sigma\}$, where σ is a threshold chosen to ensure that the selected synthetic samples are sufficiently distinct from any of the enrollment data x_{enroll}^k . This constraint helps the model better differentiate between target and unknown speakers, enhancing the overall robustness of the system. D_{syn} is the combination of all the above parts.

3.3. Tuning with outliers focusing loss functions

In fine-tuning, to focus on target variations while maintaining proper attention to outliers, we introduce an open-set training approach that models "otherness" (Figure 1). The core of this approach is the use of reciprocal points to model the "otherness" of speakers. Let $\mathbf{R}^k$ represent the reciprocal points for speaker k . The goal is to maximize the distance from $\mathbf{R}^k$.

For each speaker's enhanced enrollment data $D_{enroll+}$ (including synthetic unknown speakers), training is conducted using the speaker reciprocal point loss. The learning loss combines reciprocal points loss and prototype loss, for speaker k :

$$L_{rpl} = -\log \frac{\exp(-emb^k \cdot \mathbf{R}^k)}{\sum_{k=1}^K \exp(-emb^k \cdot \mathbf{R}^k)} + \max(\|emb^k - \mathbf{R}^k\| - radius, 0) \quad (3)$$

$$L_{proto} = -\log \frac{\exp(emb^k \cdot \mathbf{C}^k)}{\sum_{k=1}^K \exp(emb^k \cdot \mathbf{C}^k)} \quad (4)$$

where $\mathbf{C}^k$ represents the prototype center points (the mean of each target's embeddings). $\mathbf{R}^k$ and $radius$ are learned parameters. The combination of these losses results in the proposed

training loss for SpeakerRPL:

$$L_{\text{SpeakerRPL}} = L_{\text{rpl}} + L_{\text{proto}} \quad (5)$$

Other comparing training targets include prototype loss [23], normal loss [24], and angular-based softmax loss [25, 26]. We also refer to [27] for adapting the open-set logitnorm-based loss. Notably, the learning losses of SpeakerRPL and logitnorm are both outlier auto-focusing losses, meaning they do not require special handling for outliers, making them easier to implement.

4. Experiments

4.1. Experimental setting

4.1.1. Datasets and benchmarks (Table 1)

- **ESD:** It contains 10 English and 10 Mandarin speakers, with 350 utterances for each emotion per speaker across 5 distinct emotional settings [28]. The data is relatively clean for each speaker. Enrollment samples are all selected from the neutral state and testing ones from 4 emotional states (happiness, anger, sadness and surprise).
- **IEMOCAP:** It includes 10 speakers with more pronounced emotional variation based on dramatic scripts or free expressions [29]. We choose the script session which includes 5 target emotions. Each speaker has 500 utterance on average, covering all types of variations. The dataset selected from plays is relatively noisy and emotional, capturing more speaker intrinsic shifts. As in the ESD setup, neutral speech are only for enrollment, and emotional ones for testing.
- **VoxCeleb2:** Sourced from Youtube, this dataset includes a broad range of real-world samples of various speakers. The training part is following the Voxwatch [3] settings for focus on target speaker group OSI. The watch list is setting to 5 for smart home (and also for all other datasets).
- **SpeakerAging:** A self-collected dataset¹, focusing on time-varying effects, with data sourced from both YouTube and Bilibili. We select a subset of 15 speakers exhibiting significant short-term aging variation. Each speaker contributes over 1500 samples, with a balanced distribution across all weekly time spans. Only enrolled at 0 month, the testing set is evaluated at four time spans: 0, 4, 8, and 12 months.

All data, evaluation benchmarks and code are open-sourced¹.

4.1.2. Core Learning Settings

The training set includes 3 enrollment samples for 5 target speakers (5-watchlist [3]) and K-splits to cover all target speakers in the dataset, with other speakers treated as outliers. Synthetic samples are selected from 1000 sentences (similar content type) in each dataset (selecting N nearest for each, Sec 3.2), synthesized by GPTSoVITS-v2 or FishSpeech-v1.4. Additionally, the $D_{\text{enroll+}}$ set includes synthetic samples for 12 unknown speakers, selected from 200 speakers in LibriTTS [30] based on the selection policy (Section 3.2), and are treated as normal speakers. Few hyperparameters are needed: σ is set high, using the farthest synthetic unknown speakers compared to all targets. $\mathbf{R}^k$ and radius are learned, with MLP post-adaptor (the most effective tuning architecture in our experiments). Standard training settings are applied, and all other details are fully demonstrated¹.

¹<https://github.com/zhuyongchenGREAT/Few-shot-Robust-Speaker-TTS>

Table 1: The statistics of evaluation dataset.

Dataset	speaker number	target speakers trials	outliers trials	total split
Vox-O (Watchlist=40)	40	37,661	-	-
Vox2-Voxwatch [3] (Watchlist=5)	118	748,200	1,811,100	10
ESD-EN (Watchlist=5)	10	105,000	105,000	3
IEMOCAP (Watchlist=5)	10	44,085	44,085	3
SpeakerAging (Watchlist=5)	15	94,875	138,750	3

4.1.3. Metrics

The metric of this method include threshold independent metric open-set classification rate (OSCR) [21, 22], equal error rate (EER) and minDCF [3, 31]. OSCR specifically addresses open-set thresholding—distinguishing unknowns from known speakers and ensuring *correct classification* of known speakers—whereas EER/minDCF provides a broader performance measure without explicitly focusing on unknown classes.

4.2. Results

The experimental results using the proposed method are presented in Table 2 with the ERes2NetV2 foundation model [31]. In these experiments, the training target was optimized using the emotional speaker dataset [28, 29]. Without any data augmentation, direct enrollment and compare embedding (①) show unstable performance across all datasets, largely due to significant intrinsic speaker variations. Initially, we applied target speaker data augmentation directly during enrollment (②③). However, regardless of whether the controlling policy was used during training, the performance of direct enrollment degraded. For the fine-tuning-based method, we compared different sampling controlling policies with SpeakerRPL loss (starting from ④). Several observations emerged:

1. **Enrollment only:** Fine-tuning solely with 3 target enrollment data degrades performance (④).
2. **Target speaker data augmentation:** While augmenting target speaker data benefits when tuning compared to using only the enrollment, it also tends to overfit (⑤).
3. **Synthetic unknown speakers:** Incorporating unknown synthetic speakers during training is effective (⑥) and can therefore outperform the **direct** baseline (①).
4. **Targets augmentation and unknown speakers:** Training with both target data augmentation and unknown synthetic speakers (even without controlling policy) further enhances performance (⑦).
5. **Controlling unknown speakers:** Based on (⑧), controlling the unknown synthetic speakers leads to large additional improvements (⑧).
6. **Optimal Full-Controlling Policy:** The combination of controlling policy to both targets and unknown speakers yields the overall best performance (⑭).

The sampling and controlling strategy is critical across all dataset evaluations. Additionally, we compare training losses for few-shot learning (⑨–⑭), where the proposed SpeakerRPL open-set recognition loss consistently performs well, particularly with a focus on outliers, in EER, minDCF metrics, and OSCR. These findings align with the results in Table 4 using the WavLM-TDNN models [32]. Furthermore, when applying the proposed method to the SpeakerAging dataset (Table 3), we enrolled speakers only at the first time point and evaluated across subsequent time points. The proposed method consistently outperforms all baselines.

Figure 2 shows the effect of varying the number of unknown speakers and synthetic data samples (both target and unknown). While performance saturates as these numbers in-

Table 2: *Experimental results of emotional robust/wild speaker recognition based on ERes2NetV2.*

Method / Dataset	ESD (3-split eval)			IEMOCAP (3-split eval)			Vox2-Voxwatch (10-split eval)		
	EER(%)↓	minDCF(%)↓	OSCR(%)↑	EER(%)↓	minDCF(%)↓	OSCR(%)↑	EER(%)↓	minDCF(%)↓	OSCR(%)↑
① ERes2NetV2 direct enrollment only (Vox-O EER=0.57%) [31]	5.83	0.31	92.65	12.57	0.43	82.76	6.29	0.29	92.56
② +random target dataaug	6.79	0.37	91.42	14.10	0.57	79.90	9.15	0.42	87.21
③ +controlling target dataaug	7.21	0.38	90.81	14.46	0.56	79.56	8.32	0.40	88.65
④ SpeakerRPL enrollment only	8.41	0.49	80.87	12.52	0.42	74.60	8.04	0.31	87.72
⑤ +random target dataaug	6.72	0.46	81.76	11.33	0.41	75.63	10.04	0.35	84.50
⑥ +unknown (controlling unknown)	5.78	0.21	82.09	5.78	0.21	82.08	5.19	0.20	86.03
⑦ +non-controlling (random target/random unknown)	4.39	0.16	87.49	6.84	0.26	74.52	4.35	0.18	86.81
⑧ +partial-controlling (random target/controlling unknown)	2.28	0.14	92.31	5.08	0.20	80.04	4.19	0.18	87.62
⑨ Softmax full-controlling [24]	2.13	0.15	92.42	5.50	0.23	75.99	4.13	0.17	89.22
⑩ AM-Softmax full-controlling [25]	2.92	0.13	87.91	5.11	0.21	75.42	4.93	0.18	86.37
⑪ AAM-Softmax full-controlling [26]	3.69	0.15	86.01	6.06	0.23	76.89	3.66	0.16	88.97
⑫ Prototype full-controlling [23]	3.84	0.22	88.68	7.59	0.32	67.53	5.26	0.21	84.03
⑬ LogitNorm full-controlling [27]	2.40	0.17	91.02	5.43	0.23	77.66	4.11	0.17	88.59
⑭ SpeakerRPL full-controlling	2.14	0.13	93.67	4.34	0.20	84.61	2.80	0.17	93.49

Table 3: *Experimental results of time-varying robust speaker recognition based on ERes2NetV2.*

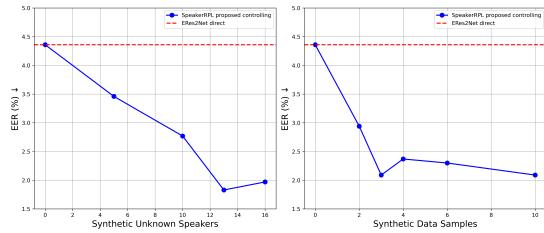
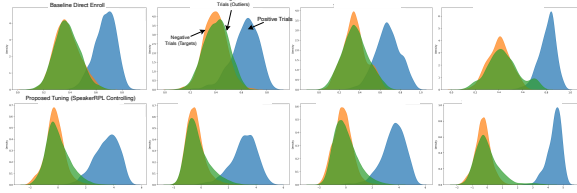
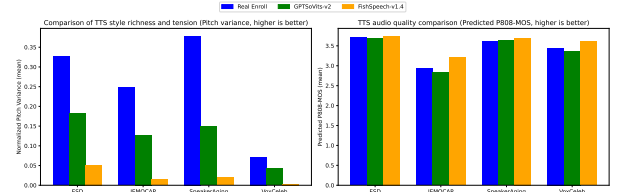
Method	0 months (enrollment/test)			4 months (test only)			8 months (test only)			12 months (test only)		
	EER(%)↓	minDCF(%)↓	OSCR(%)↑	EER(%)↓	minDCF(%)↓	OSCR(%)↑	EER(%)↓	minDCF(%)↓	OSCR(%)↑	EER(%)↓	minDCF(%)↓	OSCR(%)↑
ERes2NetV2 direct enrollment only [31]	3.60	0.20	93.13	4.84	0.39	86.34	5.33	0.37	85.28	6.79	0.54	80.20
+controlling target dataaug	4.82	0.26	92.77	5.62	0.42	85.57	6.71	0.40	84.45	7.39	0.53	80.21
Softmax full-controlling [24]	2.33	0.11	93.36	2.15	0.16	86.58	3.25	0.18	84.34	3.45	0.24	79.18
AM-Softmax full-controlling [25]	3.05	0.10	92.41	2.81	0.12	85.44	2.55	0.11	84.96	3.58	0.16	78.15
AAM-Softmax full-controlling [26]	3.84	0.13	87.32	3.84	0.17	79.20	3.92	0.17	77.86	4.81	0.21	74.18
Prototype full-controlling [23]	2.07	0.11	95.41	2.52	0.16	88.01	3.58	0.18	84.69	3.31	0.22	80.15
LogitNorm full-controlling [27]	1.69	0.11	94.72	2.08	0.15	87.04	3.02	0.16	85.24	2.83	0.20	80.58
SpeakerRPL full-controlling	1.48	0.09	95.54	1.88	0.11	88.31	2.49	0.09	87.13	2.30	0.09	81.59

Table 4: *Experimental results of emotional robust speaker recognition on WavLM-TDNN.*

Method / Dataset	ESD (3-split eval)		IEMOCAP (3-split eval)	
	EER(%)↓	minDCF(%)↓	EER(%)↓	minDCF(%)↓
WavLM-TDNN direct enrollment only (Vox-O EER=0.84%) [32]	14.81	0.89	19.12	0.74
+controlling target dataaug	23.01	0.96	19.97	0.80
SpeakerRPL enrollment only	15.72	0.95	16.67	0.87
target dataaug	16.32	0.89	18.10	0.93
non-controlling (random target/random unknown)	12.73	0.65	15.88	0.86
Softmax full-controlling [24]	12.29	0.62	14.40	0.66
Prototype full-controlling [23]	11.34	0.74	28.65	0.91
SpeakerRPL full-controlling	8.74	0.53	12.99	0.62

Table 5: *Impact of synthetic strategy comparison on ESD.*

Method	EER(%)↓	minDCF(%)↓	OSCR(%)↑
ERes2NetV2 direct enrollment only [31]	5.83	0.31	92.65
ERes2NetV2 direct +FishSpeechV1.4 [16]	8.88	0.46	87.11
ERes2NetV2 direct +GPTSoVITS [15]	6.79	0.37	91.42
SpeakerRPL +FishSpeechV1.4	3.76	0.14	88.74
SpeakerRPL +GPTSoVITS	2.14	0.13	93.67

Figure 2: *Ablation study on the influence of the number of unknown synthetic speakers and target synthetic enrollment.*Figure 3: *Score distribution comparison (zoom in for details).*Figure 4: *Comparison of TTS style and quality metrics (zoom in for details).*

crease, all increases result in improvements over the direct enrollment baseline. Figure 3 illustrates the trial score distributions, showing a distinct separation between target and outlier speakers when using the proposed method compared to the “direct” method. In Table 5, we compare different training settings with various speech synthesis models. Notably, the GPTSoVITS model in a zero-shot setting performs well, outperforming the state-of-the-art FishSpeech model. This improvement is primarily attributed to GPTSoVITS’ enhanced speech *style richness*, which effectively handles the speech variability present across all evaluation datasets. This contributed more significantly than audio quality (P808MOS) [33], which is almost identical for both TTS systems, as shown in Figure 4.

5. Conclusions

We propose a robust intrinsic speaker recognition system for open-set identification in smart home scenarios, leveraging speech synthesis and enrollment-time few-shot tuning. Our findings demonstrate that an optimal synthetic data selection policy and suitable open-set outlier-focused loss functions are critical for performance. We show that the proposed SpeakerRPL-Controlling Policy outperforms alternative approaches. Overall, the method improves training by better handling emotional and aging variations in target speakers, enhancing robustness to intrinsic speaker variability while maintaining resilience to unknown outliers. Experimental results underscore the significant value of state-of-the-art, style-rich TTS systems for speaker recognition tasks.

6. References

- [1] X. Qin, N. Li, S. Duan, and M. Li, "Investigating long-term and short-term time-varying speaker verification," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2024.
- [2] R. Tao, Z. Shi, Y. Jiang, T. Liu, and H. Li, "Voice conversion augmentation for speaker recognition on defective datasets," *arXiv preprint arXiv:2404.00863*, 2024.
- [3] R. Peri, S. O. Sadjadi, and D. Garcia-Romero, "VoxWatch: An open-set speaker recognition benchmark on VoxCeleb," Jun. 2023, arXiv:2307.00169 [cs, eess].
- [4] X. Qin, N. Li, W. Chao, D. Su, and M. Li, "Cross-age speaker verification: Learning age-invariant speaker embeddings," in *Interspeech 2022*, 2022, pp. 1436–1440.
- [5] J. Tian, X. Hu, and X. Xu, "Learning emotion-invariant speaker representations for speaker verification," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 10 611–10 615.
- [6] D. Li, Z. Yang, J. Liu, H. Yang, and Z. Wang, "Emotion embedding framework with emotional self-attention mechanism for speaker recognition," *Expert Syst. Appl.*, vol. 238, p. 122244, 2023.
- [7] F. Zhang, W. Zhou, Y. Liu, W. Geng, Y. Shan, and C. Zhang, "Disentangling age and identity with a mutual information minimization for cross-age speaker verification," in *Interspeech 2024*, 2024, pp. 3789–3793.
- [8] V. P. Singh, M. Sahidullah, and T. Kinnunen, "Speaker verification across ages: Investigating deep speaker embedding sensitivity to age mismatch in enrollment and test speech," in *Interspeech 2023*, 2023, pp. 1948–1952.
- [9] C. Du, B. Han, S. Wang, Y. Qian, and K. Yu, "Synaug: Synthesis-based data augmentation for text-dependent speaker verification," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 5844–5848.
- [10] H. Huang, X. Xiang, F. Zhao, S. Wang, and Y. Qian, "Unit selection synthesis based data augmentation for fixed phrase speaker verification," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 5849–5853.
- [11] Y. Qian, X. Gong, and H. Huang, "Layer-wise fast adaptation for end-to-end multi-accent speech recognition," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 2842–2853, 2022.
- [12] X. Qin, Y. Yang, Y. Shi, L. Yang, X. Wang, J. Wang, and M. Li, "VC-AUG: Voice Conversion Based Data Augmentation for Text-Dependent Speaker Verification," 05 2023, pp. 227–237.
- [13] Y. Zhang, Z. Li, J. Lu, W. Wang, and P. Zhang, "Synthetic speech detection based on the temporal consistency of speaker features," *IEEE Signal Processing Letters*, vol. 31, pp. 944–948, 2024.
- [14] Z. Li, M. Li, P. Zhang, Y. Ren, Z. Cai, and H. Nishizaki, "The slt 2024 source speaker tracing challenge (sstc 2024) evaluation plan,"
- [15] Rvc-Boss, "Git-Hub - RVC-Boss/GPT-SoVITS: 1 min voice data can also be used to train a good TTS model! (few shot voice cloning)." [Online]. Available: <https://github.com/RVC-Boss/GPT-SoVITS>
- [16] S. Liao, Y. Wang, T. Li, Y. Cheng, R. Zhang, R. Zhou, and Y. Xing, "Fish-speech: Leveraging large language models for advanced multilingual text-to-speech synthesis," 2024.
- [17] W.-H. Heo, J. Maeng, Y. Kang, and N. Cho, "Centroid estimation with transformer-based speaker embedder for robust target speaker extraction," in *Proc. Interspeech 2024*, 2024, pp. 4333–4337.
- [18] Q.-H. Nguyen, N. Le Minh, and M. Unoki, "Speaker verification using distance based on principal component analysis for household scenario adaptation," in *2023 RIVF International Conference on Computing and Communication Technologies (RIVF)*. IEEE, 2023, pp. 441–446.
- [19] K. C. Kishan, Z. Tan, L. Chen, M. Jin, E. Han, A. Stolcke, and C. Lee, "OpenFEAT: Improving Speaker Identification by Open-Set Few-Shot Embedding Adaptation with Transformer," in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 7062–7066, iSSN: 2379-190X.
- [20] Y. Li, H. Chen, W. Cao, Q. Huang, and Q. He, "Few-Shot Speaker Identification Using Lightweight Prototypical Network With Feature Grouping and Interaction," *IEEE Transactions on Multimedia*, pp. 1–12, 2023.
- [21] Z. Chen, Z. Ai, X. Li, and S. Xu, "Enhancing open-set speaker identification through rapid tuning with speaker reciprocal points and negative sample," in *2024 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2024, pp. 1144–1149.
- [22] G. Chen, P. Peng, X. Wang, and Y. Tian, "Adversarial Reciprocal Points Learning for Open Set Recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–1, 2021, arXiv: 2103.00953.
- [23] Y. Li, H. Chen, W. Cao, Q. Huang, and Q. He, "Few-shot speaker identification using lightweight prototypical network with feature grouping and interaction," *IEEE Transactions on Multimedia*, 2023.
- [24] Q.-B. Hong, C.-H. Wu, H.-M. Wang, and C.-L. Huang, "Combining deep embeddings of acoustic and articulatory features for speaker identification," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 7589–7593.
- [25] W. Lin and M.-W. Mak, "Robust speaker verification using deep weight space ensemble," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 802–812, 2023.
- [26] L. Zhang, Z. Chen, and Y. Qian, "Adaptive large margin fine-tuning for robust speaker verification," in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [27] H. Wei, R. Xie, H. Cheng, L. Feng, B. An, and Y. Li, "Mitigating Neural Network Overconfidence with Logit Normalization," in *Proceedings of the 39th International Conference on Machine Learning*. PMLR, Jun. 2022, pp. 23 631–23 644, iSSN: 2640-3498.
- [28] K. Zhou, B. Sisman, R. Liu, and H. Li, "Seen and unseen emotional style transfer for voice conversion with a new emotional speech dataset," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 920–924.
- [29] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, "Iemocap: Interactive emotional dyadic motion capture database," *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.
- [30] Y. Koizumi, H. Zen, S. Karita, Y. Ding, K. Yatabe, N. Morioka, M. Bacchiani, Y. Zhang, W. Han, and A. Bapna, "Libritts-r: A restored multi-speaker text-to-speech corpus," in *Interspeech 2023*, 2023, pp. 5496–5500.
- [31] Y. Chen, S. Zheng, H. Wang, L. Cheng, Q. Chen, and J. Qi, "An enhanced res2net with local and global feature fusion for speaker verification," *arXiv preprint arXiv:2305.12838*, 2023.
- [32] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing," *arXiv:2110.13900 [cs, eess]*, Jan. 2022, arXiv: 2110.13900.
- [33] C. K. Reddy, V. Gopal, and R. Cutler, "Dnsmos: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 6493–6497.